

MENA, AR, USA

WMO: 720172

Lat: 34.545N

Lon: 94.203W

Elev: 329

StdP: 97.44

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-3.0	-13.9	1.2	-0.4	-11.4	1.5	2.0	9.3	10.1	8.2	9.6	1.1	0	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	37.9	22.9	37.1	22.9	35.2	23.6	27.1	30.0	26.2	30.1	25.6	30.4	2.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	22.7	28.2	25.2	21.2	28.0	24.2	19.9	27.6	87.7	29.5	83.8	28.7	80.8	30.6	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.2	5.3	-10.1	40.1	3.0	2.1	-12.2	41.6	-13.9	42.8	-15.6	44.0	-17.7	45.5
			-10.9	27.6	2.7	1.6	-12.8	28.7	-14.4	29.7	-15.9	30.6	-17.8	31.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	6.3	8.3	13.0	17.3	21.5	26.6	28.5	27.9	24.0	17.6	12.0	7.4		
	DBStd	9.00	5.55	5.60	5.45	4.54	4.05	2.57	2.64	3.09	3.51	4.79	5.20	5.02		
	HDD10.0	419	142	91	31	4	1	0	0	0	0	2	37	112		
	HDD18.3	1542	375	282	182	72	18	0	0	0	2	73	200	339		
	CDD10.0	3186	27	45	122	223	357	498	573	556	419	238	96	32		
	CDD18.3	1267	2	2	15	41	116	248	314	298	171	49	9	1		
	CDH23.3	14356	1	9	96	359	1111	2909	3922	3577	1797	524	47	3		
	CDH26.7	6947	0	0	9	64	341	1421	2156	1974	825	153	4	0		
(14) Wind	WSAvg	1.8	2.2	2.3	2.4	2.4	1.8	1.5	1.4	1.2	1.1	1.4	1.8	1.9		
(15) Precipitation	PrecAvg	1445	84	94	137	138	169	123	122	68	122	139	136	122		
	PrecMax	2150	169	229	325	345	459	354	315	176	304	424	346	362		
	PrecMin	801	8	16	33	9	29	7	5	8	1	0	23	9		
	PrecStd	292	47	46	66	74	89	73	75	39	76	105	83	79		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	25.0	27.7	30.9	33.0	37.6	40.2	41.2	37.5	32.4	27.4	23.0		
		MCWB	14.8	17.6	18.1	19.2	22.8	21.9	21.8	22.6	21.7	21.6	19.5	18.2		
	2%	DB	20.2	22.4	26.1	27.9	32.0	36.1	37.9	38.6	35.1	30.8	24.8	22.0		
		MCWB	16.2	17.4	17.2	18.8	22.3	23.0	22.8	23.2	22.3	21.6	18.8	18.1		
	5%	DB	18.8	21.0	23.9	27.1	30.1	34.8	37.1	37.2	32.8	27.9	22.6	19.8		
		MCWB	16.3	17.3	16.9	18.5	21.7	23.4	23.1	23.2	22.6	20.0	17.6	16.3		
	10%	DB	16.4	18.0	22.3	25.9	28.0	32.7	35.1	35.0	32.0	26.3	21.2	17.1		
		MCWB	10.6	14.0	16.1	18.3	21.5	23.5	23.9	23.6	22.5	19.6	16.7	12.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	21.2	20.8	23.4	26.9	28.5	27.7	27.5	26.3	24.8	22.9	21.0		
		MCDB	20.1	22.2	23.4	25.6	29.1	31.0	30.6	30.1	29.5	28.1	24.8	22.1		
	2%	WB	18.0	19.6	19.6	21.8	25.0	27.2	26.7	26.5	25.3	23.8	20.9	19.0		
		MCDB	19.4	21.6	23.3	25.7	28.2	29.2	30.9	30.0	29.5	27.8	22.8	21.0		
	5%	WB	15.8	17.5	18.5	20.9	23.8	26.0	26.0	25.6	24.7	22.8	19.1	16.3		
		MCDB	18.0	19.2	22.8	24.3	27.6	29.5	31.0	30.8	29.2	26.5	21.7	18.3		
	10%	WB	12.1	15.3	17.4	19.9	22.9	25.1	25.4	25.1	23.9	21.4	17.3	13.5		
		MCDB	14.3	17.8	20.9	24.0	27.0	30.1	30.9	31.0	28.6	24.4	19.8	15.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.0	12.1	12.4	11.3	11.6	12.1	12.5	12.8	13.3	12.2	11.0		
		MCDBR	13.7	13.5	13.3	13.2	12.1	13.9	14.1	14.8	14.8	14.8	12.8	14.3		
	5% WB	MCWBR	9.6	8.8	7.1	6.4	5.0	4.1	3.3	3.5	4.9	6.6	8.0	11.1		
		MCWBR	11.3	11.2	11.3	10.0	9.9	10.1	11.0	11.6	11.1	10.6	9.6	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.318	0.350	0.382	0.411	0.438	0.454	0.453	0.391	0.347	0.325	0.306		
	taud		2.510	2.500	2.432	2.344	2.320	2.278	2.269	2.254	2.396	2.501	2.519	2.543		
	Ebn at Noon		898	932	924	904	877	849	832	827	869	886	876	886		
	Edh at Noon		83	93	108	124	129	134	135	135	111	92	82	76		
(44) All-Sky Solar Radiation	RadAvg		2.56	2.95	4.09	5.07	5.54	6.27	6.20	5.74	4.89	3.77	2.70	2.21		
	RadStd		0.23	0.43	0.29	0.30	0.52	0.64	0.39	0.42	0.41	0.54	0.33	0.28		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page